

Evaluating How Desktop Companion Robot Behaviors Influence Work Experience and Robot Perception

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ABSTRACT

Desktop companion robots have attracted increasing attention, yet their application to support office workers requires further exploration. In envisioning the role of desktop companion robots as a helpful “peer” for office workers, we conducted a lab experiment (N=36) to examine how desktop companion robots’ behavior influenced office workers’ productivity, work experience, and robot perception. Participants performed a digitization task alongside robots under three conditions: (a) *static* robot, displaying no interactive behaviors; (b) *work-along* robot, mirroring the participants’ work/rest patterns; and (c) *work-opposite* robot, active during participants’ rest times and inactive during work. Results demonstrated that each robot condition led to distinct perceptions of the robot (e.g., as a pet, workmate, or supervisor) and peer effects (i.e., peer presence, peer support, and peer pressure), significantly influencing participants’ task experience. Our findings offer design implications for designing future desktop companion robots to better support office workers.

CCS CONCEPTS

• **Human-centered computing** → *Empirical studies in HCI*.

KEYWORDS

Desktop companion robot, productivity, work experience, peer effect, workplace

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1 RESEARCH BACKGROUND

Companion robots, designed to offer assistance while exhibiting social behaviors [22, 28], have broadened their applications beyond their initial focus on education [8, 30], healthcare [2, 3, 42], and older adults support [9, 25]. In particular, desktop companion robots are attracting considerable interest for supporting knowledge work [20, 31]. Emerging examples, such as Eilik [14] and Cozmo [11], have entered the market, indicating a shift towards the integration of companion robots into professional work settings. In the contemporary workplace, productivity is increasingly recognized not just by the quantity of work produced but also by the quality of work and the emotional well-being of workers [23, 35, 38]. Taking this comprehensive view of productivity, desktop companion robots have the potential to foster more engaged and productive work environments by offering psychosocial and productivity support.

Recent investigations into the potential role of desktop companion robots in work settings have uncovered their potential to significantly impact work habits and overall work experience. Prior studies have specifically focused on the use of these robots to facilitate a balance between work and breaks for office workers, with interventions designed to encourage regular breaks [34, 37, 46]. Additionally, Hu and colleagues [21] developed companion robots aimed at reducing smartphone usage during work hours. Further research introduced robots that reminded users to hydrate regularly or tidy up their desks [18, 29].

The majority of these prior applications of desktop companion robots typically involve explicit interventions, like alarms or vibrations, to periodically interrupt the workflow of office workers and encourage healthy behaviors like regular breaks or hydration. Despite these advances, research on **how desktop companion robots can provide continuous, nuanced support throughout the entire work process** remains limited. Considering humans often view robots as social beings with distinct personalities, intentions, and behaviors [12, 39, 43], we envision the role of desktop companion robots as a “peer,” positioned alongside office workers to subtly and seamlessly enhance their productivity and overall work experience.

In conceptualizing such roles of desktop companion robots for office workers, the “**peer effect**” can be a useful lens. Peer effect refers to a phenomenon in which an individual’s behaviors and

attitudes are influenced by the presence of other beings [10, 16, 17, 44]. The peer effect can lead to positive outcomes, known as *peer support*, which fosters a supportive environment [24, 27]. However, it can also result in negative impacts, such as *peer pressure*, where the influence may lead to less desirable actions or feelings [10, 16, 27, 32].

As such, the influence exerted by surrounding peers could play a pivotal role in shaping productivity and the overall work experience. Interestingly, the mere existence of others, whether they are friends, strangers, or even robots exhibiting human-like characteristics, could impact one's cognitive functions and performance in both beneficial and detrimental ways [31, 36, 40, 45]. Specifically, peers whose actions are in alignment with the individual's activities could positively affect productivity, as demonstrated in studies where productivity was elevated when working in concert with others [15, 19, 44]. In contrast, the presence of peers engaging in opposite behaviors, such as continuing to work while others take breaks, could discourage individuals from engaging in healthy behaviors that could enhance their work experience [5].

To examine the peer effects of desktop companion robots on the productivity and work experience of office workers, we conducted a controlled lab experiment comparing three types of robot peers that exhibited (1) **no particular behaviors**, staying nonresponsive to the participant's state of work, (2) **similar behaviors** by working and resting concurrently with the participant, and (3) **opposite behaviors** by working when the participant is taking a break and vice versa. This experimental setup was designed to explore our primary central research question: **How should desktop companion robots be designed as "peers" to enhance office workers' productivity and work experience?**

2 METHODS

A controlled lab experiment was conducted with 36 participants. The participants performed a digitizing task under the presence of a desktop companion robot that they were randomly assigned to:

- **Static condition (N=12):** The robot exhibited no particular behaviors by displaying white light at a constant oscillating rate regardless of the task state of the participant.

- **Work-along condition (N=12):** The robot exhibited synchronous behaviors by turning on a bright blue light when the participant was active on task and displaying a white oscillating light while not on task.
- **Work-opposite condition (N=12):** The robot exhibited opposite behaviors by turning on a bright blue light when the participant was not on task and displaying a white oscillating light while active on task.

Informed by multiple pilot tests, this experiment setup was carefully developed. In particular, we found that introducing an unfamiliar robot into the workplace setting alone could draw the participants' attention away from their tasks. Furthermore, we observed that excessive movement or interaction from the robot could risk further distracting participants and impairing overall performance. We simplified the robot's actions to simply change the color of its ring-shaped LED lamp on its head.

2.1 Participants

37 participants were recruited (20 male, 17 female, age=23.9 (SD=2.8)) from the campus of a large research university via online announcements. Those who reported engaging in knowledge work alone at least 10 hours a week were selected from the applicant pool. Technical difficulties during the experiment led to the removal of data for one participant.

2.2 Experimental Setup

2.2.1 Environment. The experiment was conducted in a usability testing room, divided into the task room, where participants performed the task, and the monitoring room, where the researchers monitored and operated the experiment. Two rooms were connected via Zoom and remote desktop software. The participant was given a desktop computer (an all-in-one Windows PC with a 27-inch screen) to carry out the task, surrounded by several physical means to get rest, including comic books, a pillow, and snacks (see Figure 1-(a)). Further, the left half of the screen was occupied by a Microsoft Word application on which participants performed the digitization task, while the right half was occupied by a Google Chrome window on which participants could visit websites they liked or play online games (e.g., Tetris) (see Figure 1-(b)).

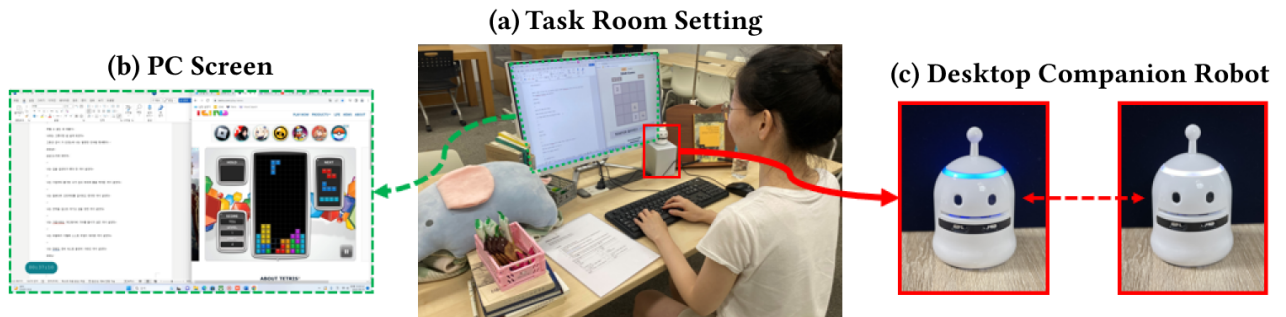


Figure 1: The experimental setup. (a) the task room setting, (b) the computer screen of a participant performing the task: the digitization task on the left half of the screen, the 'Tetris' game on the right half, (c) the color of TrueTruebot's LED lamp changes based on the participant's task state: the left shows a bright blue light, the right shows a white oscillating light.

2.2.2 Robot. “TrueTruebot” [41] (see Figure 1-(c)) was utilized due to its compact size, friendly appearance, and customizability via Python-based SDK. It possesses a diverse set of behaviors, from basic movements (i.e., moving forward, rotating) to altering the color of the ring-shaped LED lamp situated on its head, using an RGB scale from 0 to 255. In the experiment, we remotely controlled the color of the LED lamp via Bluetooth (Wizard of Oz method).

2.2.3 Task. Participants digitized the physical book’s contents (a series of poems) into a Microsoft Word document according to specific guidelines, which included detailed instructions on page indicators, paragraph structure, and word spacing. We ensured the task was simple enough to accommodate the varying capabilities of participants and repetitive enough to induce fatigue and encourage participants to take breaks. Before starting the task, participants were informed that there was no required amount of work to accomplish and that they could take breaks anytime.

2.3 Procedure

The experiment was led by two researchers: the facilitator, responsible for guiding the participants and conducting interviews, and the technician, who remotely controlled the robot from the monitoring room. All study materials and procedures were approved by the Institutional Review Board (IRB) of the university where the study was hosted. The entire experiment process was video and screen-recorded with participants’ consent. Overall, the experiments took about 2 hours.

- **Warm-up:** After obtaining written consent and introducing the overall process, the facilitator went through the task guideline booklet with participants in detail.
- **Session 1:** To let participants familiarize themselves with the lab settings and the task, participants performed the task without the robot for 25 minutes while taking breaks as he/she wished. A short break was given after Session 1.

- **Session 2:** To reduce novelty effects and highlight the robot’s interactive capabilities, the facilitator introduced the robot to participants and guided them to interact with it (i.e., waving hands to receive a greeting motion from the robot). Participants were then informed that interaction with the robot would be limited during the task. Additionally, we ensured that participants in the work-along and work-opposite conditions accurately perceived the bright blue light and the white oscillating light as indicating the robot’s active and inactive states, respectively. Next, they were instructed to repeat the same task for 25 minutes while accompanied by their assigned desktop companion robot. While then, the technician monitored the participant’s computer screen in real-time and remotely adjusted the robot’s LED color according to the conditions. Participants were considered not on task if there were no keyboard/mouse inputs for more than 3 seconds in the task window. As soon as participants made a keyboard/mouse input in the task window, they were considered to be active at work.

- **Wrap-up:** The participants filled out the questionnaire, and the facilitator conducted a brief interview about their overall experience working with robots for about 15 minutes. The participant received 20,000 Korean Won (equivalent to \$15 USD).

2.4 Measurements

The participants’ (1) perception of the robot, (2) task experience, and (3) task performance were measured in Session 2. The questionnaire used 7-point Likert scales except for robot likeability, which used 7-point semantic differential scales. All items of each dimension were averaged for analysis. A comprehensive summary of the measures can be found in Table 1.

Table 1: Summary of Measurements

Dimension	Example Item	Questionnaire	Cronbach’s α
(1) Robot Perception			
Robot Presence	<i>I was often aware of the robot in the room.</i>	Networked Minds Social Presence Inventory [6]	.83
Robot Interactivity	<i>I was able to understand what the robot meant.</i>		.88
Robot Influence	<i>My actions were influenced by the robot’s actions.</i>		.83
Robot Productivity Help	<i>The robot helped increase my productivity.</i>	Perceived usefulness scale [13]	.88
Robot Likeability	<i>Dislike - Like</i>	Godspeed Questionnaire [4]	.88
(2) Task Experience			
Perceived Productivity	<i>I felt satisfied with what I accomplished.</i>	Scales from prior research [26]	.86
Task Immersion	<i>Time appeared to go by very quickly during the task.</i>	Cognitive Absorption Scale [1]	.87
Task Enjoyment	<i>I enjoyed doing this task very much.</i>	Intrinsic Motivation Inventory [33]	.90
(3) Task Performance			
Total time spent on digitization task in Session 2 (in seconds).			
Number of characters inputted in Session 2.			
Error rate (percentage of inaccurately typed characters relative to the total character count).			

2.5 Interview Protocol and analysis

At the end of the experiment, we conducted an in-depth interview to gain a deeper understanding of participants' experience with the robot (e.g., 'How did you feel about the robot?', 'Did the robot influence your actions in any way (and vice versa)?'). Interviews were audio-recorded, fully transcribed, and analyzed using a thematic analysis method [7].

3 FINDINGS

3.1 Perception of the Robot

A series of one-way ANOVA tests was conducted to assess robot perception measures. Significant difference was identified in all measures, including robot presence ($F(2, 33)=12.64, p=.001$), interactivity ($F(2, 33)=18.44, p<.001$), influence ($F(2, 33)=36.81, p<.001$), productivity help ($F(2, 33)=10.15, p=.003$), and likeability ($F(2, 33)=3.09, p=.05$). Post hoc Tukey HSD indicated that the robots in the work-along and work-opposite conditions were considered to possess greater presence and be more interactive, leading participants to perceive that the robots had a greater influence on them during tasks and improved their productivity, compared to that in the static condition ($p<.05$ respectively). No significant difference was found between the work-along and the static conditions ($p>.05$ respectively). The only exception was observed in robot likeability measures, with participants expressing significantly less liking for the work-opposite robot ($M=4.1, SE=.30$) compared to those in the work-along ($M=5.00, SE=.30; p=.04$) and static conditions ($M=5.05, SE=.30; p=.03$). There was no significant difference between the work-along and static conditions ($p=.90$).

3.1.1 Insights from the Interview. The in-depth interview revealed how participants perceived the robot differently in each condition. Since the **static** robot emitted white oscillating light regardless of the participant's task state, the majority agreed that the robot exhibited little to no perceived interaction, resulting in a low level of presence and influence (P26, P32). Therefore, the static robot was likened to "*a puppy that quietly looks at me*" (P2), which participants perceived as not particularly helpful in enhancing productivity compared to the other two conditions.

In the **work-along** condition, participants perceived the robot's behavior mirroring their own task state as indicative of the robot working alongside them (P7, P36). Consequently, the robot was seen as engaging in more interactions and taking on a more proactive role in assisting participants with their work tasks. As a result, participants characterized the work-along robot as "*a studymate*" (P6) or "*a friend helping me complete the task*" (P7).

Conversely, participants perceived the contradictory behaviors of the **work-opposite** robot as closely monitoring and evaluating their work progress, characterizing it as "*a surveillant*" (P13, P18, P33) or "*a supervisor who assesses my work*" (P5, P15, P29). Hence, the work-opposite robot not only exhibited a heightened sense of presence but also exerted significant authority over participants' task processes, such as demanding a greater workload completion without taking breaks (P5, P29). This perceived authority of the work-opposite robot led participants to express discontent towards the robot, resulting in its lower likability compared to the other two conditions.

3.2 Task Experience

A series of one-way ANOVA analyses were conducted to compare the effects of robot conditions on task experience. There was no significant difference in perceived productivity ($p=.14$). However, a significant effect of robot conditions was observed on task immersion ($F(2, 33)=3.45, p=.04$). Post hoc Tukey HSD tests showed that participants in the work-opposite condition ($M=4.22, SE=.36$) experienced significantly lower task immersion compared to those in the work-along ($M=5.44, SE=.36; p=.02$) and static conditions ($M=5.33, SE=.36; p=.03$). No significant difference between work-along and static conditions ($p=.83$). Additionally, the robot condition had a significant impact on task enjoyment ($F(2, 33)=4.76, p=.01$). Mirroring the findings on task immersion, participants in the work-opposite condition ($M=3.83, SE=.30$) reported significantly lower task enjoyment compared to those in the work-along ($M=5.09, SE=.30; p<.01$) and static conditions ($M=4.69, SE=.30; p=.03$). No significant difference between work-along and static conditions ($p=.49$).

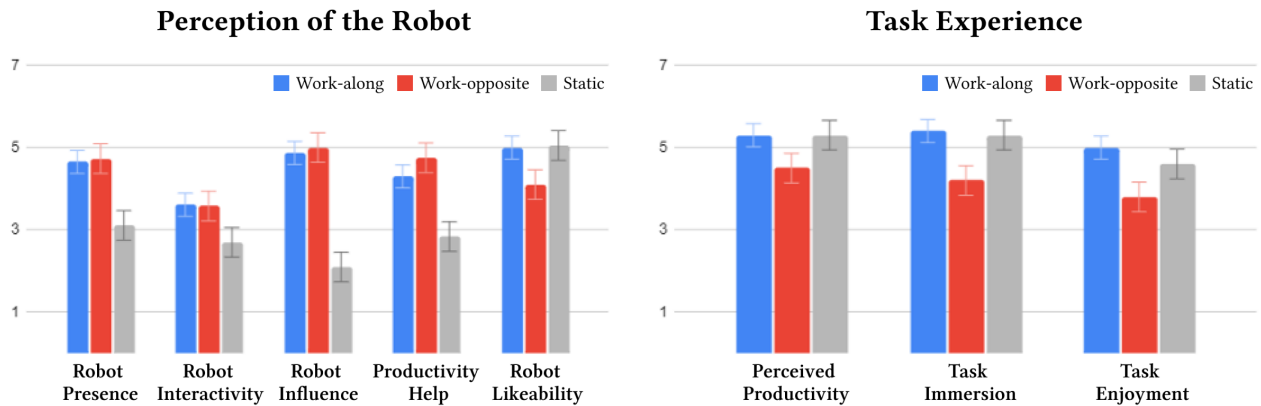


Figure 2: The effect of three types of desktop companion robots on task experience and perception of the robot.

3.2.1 Insights from the Interview. Participants' task experience varied across three conditions, closely tied to their perception of the robot. Even with minimal interaction, some participants reported that the mere presence of the **static** robot transformed the otherwise dull work environment into a more friendly space (P2, P12, P27), positively shaping their task experience. They described how the subtle presence of the robot fostered "**a comfortable and relaxed atmosphere**" (P2, 8), while the gently wavering light provided "a calming background" (P22). Also, some mentioned this resulted in "**a focused work environment**" which resembled "*a library with the subdued sound of pages turning*" (P37).

Similar to the static condition, some mentioned that simply having the **work-along** robot next to them evoked a sense of "**co-presence**" (P28), alleviating feelings of loneliness and boredom and thereby contributing to a positive task experience (P6, P11). In addition, since the work-along robot switched its color synchronously to the participants' work state, many perceived the robot as working simultaneously alongside them, which offered encouragement as if saying, "*You're doing great!*" (P7, P36) and created a **supportive work environment** (P20, P28).

In contrast, participants highlighted that the **work-opposite** robot induced a **sense of pressure**, leading to anxiety and irritation among them (P5, P33). Particularly, P15 even reported physiological distress, noting, "*I was tapping my foot constantly, my face was stiff, and my heart was pounding fast.*" Moreover, the robot's perceived role as a supervisor contributed to an unpleasant experience, with participants feeling "*judged unfairly by its standards*" (P15) and criticized as though they were being told, "*You're not doing it right!*" (P13). As a result, interaction with the work-opposite robot might have significantly reduced participants' levels of task immersion and enjoyment. Nevertheless, the robot's behavior, perceived as monitoring, was seen as beneficial for enhancing task performance, leading to perceived productivity comparable to those in other robot conditions.

3.3 Task Performance

A series of one-way ANOVA analyses were conducted to examine differences in task performance across three robot conditions. The results revealed no statistically significant differences in all task performance measures: time spent on the task (work-along: 1142 sec, work-opposite: 1234 sec, static: 1246 sec; $p = .39$), quantity of characters typed (work-along: 3062, work-opposite: 2837.1, static: 3135.2; $p = .60$), and error rates (work-along: 0.012, work-opposite: 0.017, static: 0.007; $p = .20$).

3.3.1 Insights from the Interview. Although the actual task performance measures did not vary between the robot conditions, we examined the participants' views on the robot's impact on their task performance.

In the **static** condition, opinions were divided. Some viewed the static robot as a passive spectator, **barely affecting their work performance** (P21, P22), while others felt it supported their work by creating a distraction-free, comfortable environment (P12, P25, P27). In particular, P8 found the static robot could be helpful for extended work periods, comparing its effect to the peer support of her sister's presence: "*Just like how my sister's presence during study*

times helps me stay focused, I think the robot could serve a similar role."

Many participants reported that the behavior of the **work-along** robot, which was in sync with the work states of the participants, created an **atmosphere of collaboration**. This led to feelings of encouragement, as highlighted by P36: "*Seeing the robot's color synchronized with my work progress cheered me up.*" Other participants also commented that the work-along robot facilitated accomplishing more work (P20) and avoiding distractions (P36). Given its supportive role, P11 further envisioned using the work-along robot in situations where "*one feels lonely and exhausted, especially late at night*" (P11).

Finally, in the **work-opposite** condition, participants found the robot's behavior, which was contradictory to their work state, to be a **forceful nudge**, pressuring them to reduce distractions and focus on their work (P5, P18, P29). Furthermore, such nature of the work-opposite robot motivated participants to put forth more effort by "*continuously engaging in the task without taking breaks*" (P13). While participants perceived that the authoritative presence of the work-opposite robot enhanced the quantitative output of their work, they did not see it necessarily improving the quality of their work. Therefore, they recommended its use for simple, repetitive tasks where work output volume was prioritized over quality (P15, P29, P30).

4 DISCUSSION

Our findings revealed that the robot's behavior critically affected its perception by participants, thereby shaping participants' task experiences and influencing perceived task performance through distinct peer effect mechanisms in each experimental setup.

The **static robot**, characterized by its white oscillating light that was independent of the participants' work states, was perceived as a 'quiet puppy' offering silent companionship without disrupting work activities. This robot's minimal interaction and subtle presence led some participants to report a negligible influence on their productivity. Nevertheless, they appreciated the static robot's **peer presence**, noting its non-intrusive, calming nature, which fostered a comfortable environment for concentration and task engagement.

The **work-along robot**'s behavior that mirrored the participants' work states was seen as collaborative, enhancing the level of interaction and perceived presence. Described as a 'helpful workmate,' the work-along robot was praised for enriching the task experience by offering **peer support**, encapsulating co-presence and encouragement. This supportive environment was believed to enable participants to increase their productivity and minimize distractions.

Conversely, the **work-opposite robot**, with its behavior contrasting participants' work states, was viewed as having an authoritative and evaluative presence. Therefore, participants illustrated the robot as a 'supervisor' who consistently monitored and evaluated their work progress. On the one hand, this approach was seen as beneficial for increasing work quantity by regulating distracting behaviors and discouraging breaks. However, it also introduced significant **peer pressure**, leading to less enjoyable task experiences.

Overall, our findings provide useful insights for the design of desktop companion robots, emphasizing the use of distinct peer

effects through the robots' behaviors to create unique personalities and relationships with human workers. In particular, we suggest employing the static behavior to foster calming environments with minimal distractions and a subtle presence, the work-along behavior for situations that need encouragement, and the work-opposite behavior for tasks requiring urgency or a focus on output quantity. Nonetheless, caution should be exercised when employing the work-opposite behavior due to its potential to create stress and provoke adverse reactions from users.

Interestingly, the majority of the existing design for desktop companion robots has centered around principles akin to work-opposite behaviors. These designs typically prompt users to undertake actions they are not currently engaged in (e.g., taking a break while working [34, 37, 46]) or to cease certain activities (e.g., stopping phone use [21]). Future explorations in desktop companion robot design could significantly benefit from investigating the effects of static and work-along behaviors. This shift might lead to the development of more nuanced and positively received desktop companion robots that cater effectively to the diverse needs of users in their work environments.

One notable inconsistency of our findings was the absence of significant differences in task performance among the three robot conditions, despite variations in participants' perceptions toward the robot behaviors and their productivity. This discrepancy might stem from the constraints of our controlled lab experiment settings, which featured limited methods of interaction and engagement duration. As a result, it was likely that the full potential effects of each robot's behavior had not been reflected in measurable work outcomes. To address this gap, conducting long-term, in-the-wild studies that incorporate a broader spectrum of robot interaction methods and longer interaction periods would be essential.

5 CONCLUSION

In this paper, we explored the impact of desktop companion robots' behaviors on the productivity and work experience of office workers through a controlled lab experiment. The results revealed that different robot behaviors significantly affected how participants perceived the robot, shaping task experience and influencing perceived task performance through unique peer effect mechanisms. In particular, with minimal interaction and subtle presence, the static robot behavior was seen as assisting the participants in performing the task in a more focused and comfortable environment. In the work-along condition, the robot was perceived to provide encouragement by creating a sense of working together. In contrast, the work-opposite behavior, while prompting enhanced concentration and productivity, led to unpleasant task experiences due to excessive pressure and authority imposed on the participants. Based on our findings, we provided design implications for future desktop companion robots to support office workers.

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